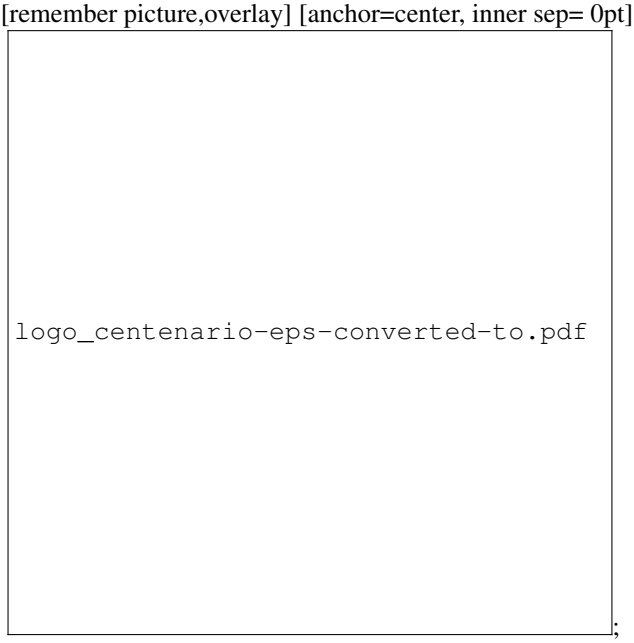
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UNIVERSIT DEGLI STUDI DI TRIESTE

DIPARTIMENTO DI SCIENZE ECONOMICHE, AZIENDALI, MATEMATICHE E STATISTICHE - DEAMS

Corso di Laurea in Quello Che Vi Pare a Voi

TESI DI LAUREA TRIENNALE

DERIVATIVES PRICING AND STRATEGIES: A FOCUS ON AGRICULTURAL COMMODITIES

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ANNO ACCADEMICO 2023–2024

A Francesca



Figure 1: Comparison between low variance (left) and high variance (right) Geometric Brownian Motion.

Commodity	Product Code	Price Quotation	Contract size
Class III Milk	DC/DA	U.S. dollars and cents per cwt	2,000 cwt
Class IV Milk	GDK/DK	U.S. dollars and cents per cwt	2,000 cwt
Non-fat Dry Milk	GNF	U.S. cents per pound	44,000 lb
Dry Whey	DY	U.S. cents per pound	44,000 lb
Cash-settled Butter	CB	U.S. cents per pound	20,000 lb
Cash-Settled Cheese	CSC	U.S. dollars and cents per pound	20,000 lb

Table 1: Dairy.
Author s elaborations based on data from CME group.

Introduction This chapter is on how to upload an image and refer to it. See figure 1. Of course you gotta add some text otherwise the page will be empty and the figure will be automatically placed at the top or at the bottom of the page.

1 This is how to upload a figure

1.1 And refer to a figure

How to create a table

2 How to cite a table

This is how to create a table, and cite a table. See table 1.

3 Now that you see the table

3.1 You can copy it

Of course you gotta add some text otherwise the page will be empty and the table will be automatically placed at the top or at the bottom of the page.

I sette nani

This chapter is completely useless.

4 Mammolo Eolo

4.1 Pisolo

4.1.1 Non me li ricordo

4.2 Brontolo

5 Stronzolo

6 Jeremy Clarkson

7 James May

8 Richard Hammond

Other Stuff

9 How to draw a binomial tree.

This chapter shows how you can create a recombiant binomial tree.

[>=stealth,sloped] (tree) [matrix of nodes, minimum size=0.3cm, column sep=0.3cm, row sep=0.3cm,]
Su^3
Su^2
Su Su^2d
S Sud ...
Sd Sud^2
Sd^2
Sd^3

; [->] (tree-5-1) – (tree-4-2) node [midway,above] ; [->] (tree-5-1) – (tree-6-2) node [midway,below] ; [->] (tree-4-2) – (tree-3-3) node [midway,above] ; [->] (tree-4-2) – (tree-5-3) node [midway,below] ; [->] (tree-6-2) – (tree-5-3) node [midway,above] ; [->] (tree-6-2) – (tree-7-3) node [midway,below] ; [->] (tree-3-3) – (tree-2-4) node [midway,above] ; [->] (tree-3-3) – (tree-4-4) node [midway,below] ; [->] (tree-5-3) – (tree-4-4) node [midway,above] ; [->] (tree-5-3) – (tree-6-4) node [midway,below] ; [->] (tree-7-3) – (tree-6-4) node [midway,above] ; [->] (tree-7-3) – (tree-8-4) node [midway,below] ; [->] (tree-2-4) – (tree-1-5) node [midway,above] ; [->] (tree-2-4) – (tree-3-5) node [midway,below] ; [->] (tree-4-4) – (tree-3-5) node [midway,above] ; [->] (tree-4-4) – (tree-5-5) node [midway,below] ; [->] (tree-6-4) – (tree-5-5) node [midway,above] ; [->] (tree-6-4) – (tree-7-5) node [midway,below] ; [->] (tree-8-4) – (tree-7-5) node [midway,above] ; [->] (tree-8-4) – (tree-9-5) node [midway,below] ;

Figure 2: Markov process evolution

10 How to make a bullet list.

To better understand the pricing models let's first introduce some statistical concepts. Brownian motion W_t is a stochastic process with four main features:

- Starts at time t: $W_0 = 0$;
- W_t is continuous in time;
- W_t increments (the differences $X(t) - X(s)$ between two intervals in time t and s) are independent;

- $W_t - W_s \sim \mathcal{N}(0, s-u), (\forall 0 < s < t)$

Glossary Here you can write your glossary.

Ringraziamenti
Un grazie che non pu essere taciuto va a Luigi Magliocchi, faro nel buio di UniTs.